fy-m1: Teaching an Open Model to Chat in Everyday Malagasy

Version 0.1 — full human ratings in v0.2

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Abstract

Open language models that cover Malagasy tend to answer like an official letter: formal, stiff, and often half-translated from French or English. People in Madagascar do not text like that. We fine-tune an open model, **pending**, with QLoRA on a single free GPU so that it replies the way a friend would: short, friendly, mixing in French, and using everyday spelling. We describe how we chose the base model, how we built and anonymised the training data, how we trained and trimmed the model, and what we measured. On **pending** fresh everyday prompts, we report the share of replies that drift into full French or English (base: **pending**; fy-m1: **pending**), LLM-judge scores, and regression checks on reading and translation. A blind human comparison reports early votes here; the full panel's result comes in v0.2.

1 Motivation: the formal-versus-everyday gap

Malagasy is the national language of Madagascar, but most Malagasy text online is formal: news, Wikipedia, religious and government writing. Models trained on that text learn that register. When a user writes a short, casual message, they get back a polished paragraph that no one would send to a friend.

Everyday Malagasy looks different. French words are mixed in freely, including French verbs used with Malagasy grammar. Spelling is phonetic and shortened. Replies are short, with humour and emoji. For this project, that mixed style is the target, not an error. A reply only counts as drifting if it is entirely French or English with no Malagasy in it.

Our goal is modest and measurable: a model that is clearly better than its base model at everyday Malagasy chat, without losing much general Malagasy ability, trained and served at zero cost.

2 Choosing the base model

We shortlisted recent open-weight instruction models between 7B and 12B parameters that fit QLoRA training on a 16 GB T4 GPU. Each candidate was loaded in 4-bit and scored on: reading comprehension (Belebele, `plt_Latn`; Bandarkar et al., 2024); translation in both directions with English (FLORES+ devtest, chrF++; Goyal et al., 2022, NLLB Team et al., 2022, Burchell et al., 2024, Popović, 2017); tokenizer efficiency (tokens per Malagasy word on Wikipedia sentences); and a short everyday-chat sniff test rated by a native speaker (1–5). License was a tie-breaker: we prefer permissive licenses when quality is close. Table 1 shows the numbers. We chose **pending** (license: **pending**).

Model	License	Belebele	chrF++ mg→en	chrF++ en→mg	Tokens/word	Natural (1–5)
pending						

Table 1: Base-model candidates, all loaded in 4-bit. Higher is better except tokens per word.

3 Data

We describe the data but do not share it. It contains private conversations.

Real chats. The main source is one native speaker's Messenger export, using both sides of each conversation. We parsed **pending** threads into **pending** short multi-turn conversations, splitting on long gaps and merging back-to-back messages from the same side of the conversation. We kept conversations where at least half of the turns are Malagasy or Malagasy–French (language ID with GlotLID; Kargaran et al., 2023). A native speaker reviewed samples and approved **pending** conversations.

Formal slice. **pending** short single-turn examples built from Malagasy Wikipedia, so facts and standard spelling survive. We keep this slice small so it does not pull the style back to formal.

Identity and refusals. **pending** short exchanges, written by the native speaker, that answer "who are you?" consistently and politely refuse harmful requests.

Synthetic chats. **pending** synthetic conversations. We only planned to add them if there were too few real chats; any synthetic chats come from an open-weight model run by us, never a closed API, and a native speaker checks a sample.

In total the model trained on **pending** examples.

Privacy process. Raw exports never left the owner's laptop. A local script replaced names and nicknames (also inside message text), phone numbers, emails, links and account handles before anything was uploaded. Only the cleaned, approved data went to a private dataset repository, used by the training notebook and never made public. Test prompts were written fresh, so the test set contains no private text and is public.

4 Method

We train with QLoRA [Dettmers et al., 2023]: the base model is frozen in 4-bit NF4 and we learn LoRA adapters [Hu et al., 2022] on all linear layers, with rank **pending** and alpha **pending**. We use Unsloth [Han et al., 2023] on a free Google Colab GPU (**pending**). Settings: learning rate **pending** with a **pending** schedule, sequence length **pending**, effective batch size **pending**, **pending** epochs (**pending** steps, **pending** of training). Loss is computed on assistant turns only, in the base model's own chat template, with a fixed system prompt in Malagasy. Free sessions disconnect, so checkpoints are pushed to a private repository during training and a new session resumes from the latest one. Final training loss was **pending**; best held-out loss was **pending**.

5 Making the model smaller: vocabulary trimming

Much of a multilingual model's vocabulary is for scripts and languages fy-m1 never uses. Following Ushio et al. [2023], we keep only the tokens seen in Malagasy, French and English text, plus all special and byte-fallback tokens, and drop the matching embedding and output rows. The trim ships only if it passes three gates: held-out Malagasy text round-trips through the new tokenizer unchanged; greedy answers are unchanged for at least 95% of the test prompts; and Belebele stays within one point. Result: gates **pending**. Vocabulary **pending** → **pending** tokens; parameters **pending** → **pending**; download **pending** → **pending**; speed on the demo CPU **pending** → **pending** tokens/s.

Removing whole layers [Gromov et al., 2024] would save more, but needs healing with extra training that free GPU time cannot pay for. We leave it for v0.2.

6 Results

We generated answers from the base model and fy-m1 to **pending** fresh everyday prompts with the same system prompt and sampling settings (Table 2).

Measure	Base	fy-m1
Drift to full French/English (↓)	**pending**	**pending**
Naturalness, LLM judge, 1–5 (↑)	**pending**	**pending**
Correctness, LLM judge, 1–5 (↑)	**pending**	**pending**
Held-out chat perplexity (↓)	**pending**	**pending**
Belebele accuracy (↑)	**pending**	**pending**
FLORES+ chrF++ mg→en (↑)	**pending**	**pending**
FLORES+ chrF++ en→mg (↑)	**pending**	**pending**

Table 2: Base model versus fy-m1. Belebele and FLORES+ are regression checks: they should not fall much.

Automatic judge. An LLM judge scored both answer sets blind, with labels hidden and order shuffled, on a published 1–5 rubric for naturalness and correctness [Zheng et al., 2023]. Judge: **pending**. Judge scores are a cheap signal, not a substitute for native speakers.

Leak check. Any fy-m1 answer that shares a run of 8 words with the training chats blocks release. **pending** of fy-m1 answers had such a run; across all fy-m1 answers there were **pending** distinct matched 8-word runs. The owner reviewed each run on its own, seeing only those 8 words and never the chat around them, and release required every run to be a generic phrase. Of those runs, **pending** were judged generic and **pending** were rejected.

Blind human comparison (early votes). Native speakers rate pairs of answers in a public "arena": same prompt, two unlabelled answers in random order, pick the more natural one. These are the early votes in at release: **pending** votes from **pending** raters; fy-m1 win rate **pending** (95% Wilson interval **pending**–**pending**; Wilson, 1927). With so few raters this is a first signal, not a result. The full panel's numbers come in v0.2.

Reproducibility. The test prompts, rubric, code and notebooks are public. Version 0.1 releases only aggregate scores; the per-prompt answers of both models are released in v0.2, after the human comparison closes, so that raters cannot tell which model wrote which answer. The training chats stay private, so training can be rerun only on other data.

7 Limitations

- The style comes mostly from one person's chats and their contacts. It reflects Antananarivo-style everyday speech, not other regions or dialects.
- The human comparison has only early votes from **pending** raters. Until the full panel is in, the main evidence is automatic.
- The drift metric can be a little low. A reply counts as Malagasy if GlotLID says so, or if at least 10% of its words are common Malagasy words. English that contains a Malagasy name or a quoted Malagasy word (e.g. "Hi Vola, how are you?") can therefore pass. Fully French or English replies with no Malagasy words are counted as drift.
- An LLM judge may not understand Malagasy slang well; its scores can be wrong in either direction.
- The model is small and runs on a free CPU. It can be slow, get facts wrong, and invent things.
- We train only once or twice on free hardware, so settings are not carefully tuned.

8 Ethics, consent and privacy

The data owner confirmed that everyone in the conversations consented to their messages being used. Raw data never left the owner's machine; only anonymised, approved data was used for training, and it is not released. We check that the model does not repeat training text before release (Section 6). The model was trained with a small set of polite refusals and gets a basic safety check before release, but it is not a safety-tested product. No closed-API model outputs are in the training data.

9 What comes next (v0.2)

- The full panel's blind human comparison, with confidence intervals.
- Each model's answer to each test prompt.
- The layer-pruning experiment.
- Longer training and more everyday data.

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